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Role

Voice Agent & Complaints Handling

Tech Stack

FastAPI Python gTTS SQLite REST API (v1)

1. Voice Agent

- Worked alongside Shalini on the voice agent — the speech layer used to narrate generated video content (gTTS-based text-to-speech).
- Focused on the input/handling side: turning script text into a clean voice-ready format before it's passed to speech synthesis.
- Built `voice_agent.py`: strips stage cues, brackets, markdown, normalizes regional acronyms (MAS, PDPA, MOH, AI, B2B) and regional currencies (S\$, RM, HK\$, USD), and optimizes sentence pacing.

How it works: Content Agent produces the script → voice agent prepares and formats it → gTTS generates the audio → audio is handed to the video assembly step.

2. Complaints

- Built the complaints handling flow — capturing user/customer complaint input, storing it (SQLite), and routing it for review/response, separate from the marketing content pipeline.
- Developed automated triage and department routing: claims denial → `claims_desk`; regulatory/disclosure → `compliance_legal`; policy → `underwriting_team`; billing → `billing_support`; urgent keywords (lawsuit/regulator) → `executive_escalations`.

How it works: complaint is submitted → stored as a record in the database → flagged/routed so it can be reviewed and actioned, with a status field similar in spirit to the content queue's pending → approved pattern (pending → `under_review` → `routed` → `actioned` → `resolved/rejected`).

Implementation & File Paths

Complaints Schema (SQLite): Complaint table with `id`, `complaint_number` (CMP-YYYYMMDD-XXXX), `customer_name`, `customer_email`, `brand`, `category`, `priority`, `subject`, `description`, `status`, `routed_to`, `routing_reason`, `assigned_reviewer`, `response_draft`, `resolution_notes`, `actioned_by`, `actioned_at`, `resolved_at`.

Exact Routing Logic: Auto-triage categorizes by complaint topic and risk profile, routing to designated operational teams with manual re-assignment and resolution actions.

Key File Paths:

- `backend/app/services/voice_agent.py` — Voice script sanitization & normalization
- `backend/app/services/voice_service.py` — Speech synthesis pipeline integration
- `backend/app/models/entities.py` — Complaint SQLite ORM entity
- `backend/app/schemas/dtos.py` — Complaint DTOs & response schemas
- `backend/app/services/complaints_service.py` — Complaints routing & lifecycle engine
- `backend/app/api/v1/complaints.py` — REST API endpoints (v1)
- `backend/tests/test_voice_agent_prep.py` & `backend/tests/test_complaints_flow.py` — Automated test suites